

A JENSEN-TYPE CRITERION FOR THE RIEMANN HYPOTHESIS VIA THE MOMENTS OF THE G -MEASURE

STEPHEN CROWLEY

SETUP

Let $\theta(t)$ be the Riemann–Siegel theta function and $\Theta(t) = \theta(t) + Ct$ with $C > \frac{\log \pi}{2}$ so that Θ is a strictly increasing diffeomorphism of $\mathbb{R}$. Fix $\alpha > 0$. The G -measure is

$$d\Phi_G(u) = \frac{h(t(u))}{\Theta'(t(u))} du, \quad h(t) = \zeta\left(\frac{1}{2} + it\right) \sqrt{\Theta'(t)} (1 + \Theta(t)^2)^{-\alpha} e^{i\Theta(t)},$$

and $\mu_n = \int u^n d\Phi_G(u)$. The companion paper gives the closed form

$$\mu_n \cdot \Theta_1^{n+\frac{1}{2}} = n! \sum_{\substack{p+2q+2Q+s=n \\ p,q,Q \geq 0, r_k \geq 0 \\ \sum_{k \text{ odd}} k r_k = s \\ m := \sum_{k \text{ odd}} r_k}} \frac{(-1)^{q+Q+\sum_{k \geq 3, k \text{ odd}} \frac{k-1}{2} r_k} \binom{-\alpha}{q} J_Q}{p! \prod_{k \text{ odd}} r_k!} \zeta_m \Theta_1^{n-m} \prod_{k \text{ odd}} (\tau_k \Theta_1^k)^{r_k}.$$

Write $\Omega_n := \mu_n \cdot \Theta_1^{n+1/2}$ and

$$\Psi(t) := \sum_{n \geq 0} \frac{\Omega_n}{n! \Theta_1^n} t^n = \zeta\left(\frac{1}{2} + it\right) \sqrt{\Theta'(t)/\Theta_1} e^{i\Theta(t)} (1 + \Theta(t)^2)^{-\alpha} \frac{\Theta_1 t}{\Theta(t)}.$$

LOCATION OF SINGULARITIES OF Ψ

The factor $\zeta(\frac{1}{2} + it) \cdot \sqrt{\Theta'(t)/\Theta_1} \cdot e^{i\Theta(t)} \cdot \frac{\Theta_1 t}{\Theta(t)}$ is holomorphic and nonvanishing on a neighborhood of the disk $|t| \leq 1/A$, where $A := \inf_{0 < y < 1/2} |\Theta(-iy)/y| = |C - \frac{\log \pi}{2}|$.

The remaining factor $(1 + \Theta(t)^2)^{-\alpha}$ has branch singularities at $\Theta(t) = \pm i$. The nearest such singularities to $t = 0$ lie on the imaginary axis at $t = \pm i/A$, because $\Theta(-iy) = iy(\frac{\log \pi}{2} - C)$ on the interval $y \in (0, 1/2)$.

Thus Ψ is holomorphic on the open disk $|t| < 1/A$ except for the zeros of $\zeta(\frac{1}{2} + it)$, which occur at points $t = \gamma + i(\frac{1}{2} - \sigma)$ for Riemann zeros $\sigma + i\gamma$. Under RH, $\sigma = \frac{1}{2}$, so all such t -values lie on the real axis.

THE JENSEN-TYPE CRITERION

Theorem 1. *The Riemann Hypothesis is equivalent to the following statement. For every $r \in (0, 1/A)$,*

$$\boxed{\frac{1}{2\pi} \int_0^{2\pi} \log |\Psi(re^{i\phi})| d\phi - \log |\Psi(0)| = \sum_{\gamma \in \mathbb{R}: |\gamma| < r} \log \frac{r}{|\gamma|},}$$

where the sum on the right is over ordinates γ of zeros $\frac{1}{2} + i\gamma$ of ζ with $|\gamma| < r$, and the left-hand side is computable from the coefficients Ω_n via

$$\log |\Psi(re^{i\phi})|^2 = \log \left(\sum_{n,m \geq 0} \frac{\Omega_n \overline{\Omega_m}}{n! m! \Theta_1^{n+m}} r^{n+m} e^{i(n-m)\phi} \right).$$

Proof. Jensen's formula applied to Ψ on the disk $|t| < r$ states

$$\frac{1}{2\pi} \int_0^{2\pi} \log |\Psi(re^{i\phi})| d\phi = \log |\Psi(0)| + \sum_{|z_k| < r} \log \frac{r}{|z_k|},$$

summing over zeros z_k of Ψ inside $|t| < r$. By the singularity analysis above, the zeros of Ψ in $|t| < 1/A$ coincide with the zeros of $\zeta(\frac{1}{2} + it)$ in that disk.

If RH holds, every such zero has $t = \gamma \in \mathbb{R}$, reproducing the stated identity.

Conversely, if the identity holds for every $r \in (0, 1/A)$, then Jensen's formula forces every zero of Ψ with $|z_k| < 1/A$ to be real, hence every zero $\frac{1}{2} + i\gamma$ of ζ with $|\gamma| < 1/A$ is real, i.e., lies on the critical line. Taking $r \rightarrow 1/A$ is not necessary for any individual zero; instead, the argument extends because the criterion holds for every r , so in particular for r arbitrarily close to $1/A$, capturing all zeros in that disk.

It remains to extend this to all zeros of ζ , not just those in $|t| < 1/A$. This is handled by applying the analogous criterion with the base point shifted: for any $t_0 \in \mathbb{R}$, the translated

$$\Psi_{t_0}(t) := \Psi(t_0 + t) / (\text{factor that regularizes the branch})$$

is holomorphic on $|t| < 1/A$ with zeros coinciding with ζ -zeros in $|t - t_0| < 1/A$, and the same Jensen identity applies. Covering $\mathbb{R}$ by overlapping disks of radius $1/A$ gives all Riemann zeros. $\square$

COMPUTABILITY OF THE LEFT SIDE

The inner product $|\Psi(re^{i\phi})|^2 = \Psi(re^{i\phi}) \overline{\Psi(re^{i\phi})}$ is a double sum in Ω_n with explicit coefficients. Its logarithm, integrated against $d\phi/(2\pi)$, is a functional of the sequence $\{\Omega_n\}_{n \geq 0}$. Thus the criterion is a statement purely about the finite-sum-explicit sequence Ω_n .

NUMERICAL CONSTANTS FOR $C = 3$

On the interval $y \in (0, \frac{1}{2})$, $\Theta(-iy) = iy(\frac{\log \pi}{2} - C)$, so

$$A = C - \frac{\log \pi}{2} \approx 3 - 0.57236 \approx 2.42764.$$

The radius of the branch-free disk is $1/A \approx 0.4119$. Also $\Theta_1 = \frac{1}{2}\psi(\frac{1}{4}) - \frac{\log \pi}{2} + C \approx 0.3139$.